rsna-knee-abnormality-detection

7th place solution with no external training data

Rank	7
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Topic ID	752217
Version	Original

Source: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/discussion/752217
Retrieved with Kaggle CLI on 2026-10-27.

Only one model: public #1 at 0.969, private 0.958, with no external training data

Private LB 0.957 (7th) · Public LB 0.970 · Best single model 0.969 / 0.958 · Trained on competition data only

First of all, I would like to sincerely thank the Radiological Society of North America (RSNA), the contributing institutions and the radiologists who annotated the data, and Kaggle for organizing and hosting the RSNA Knee Abnormality Detection competition. Building a multi-site knee MRI dataset with reports in many languages and an expert-adjudicated gold set takes a great deal of work, and it made this a challenging and rewarding problem. I also thank everyone who shared ideas and findings in the discussion forum.

Context

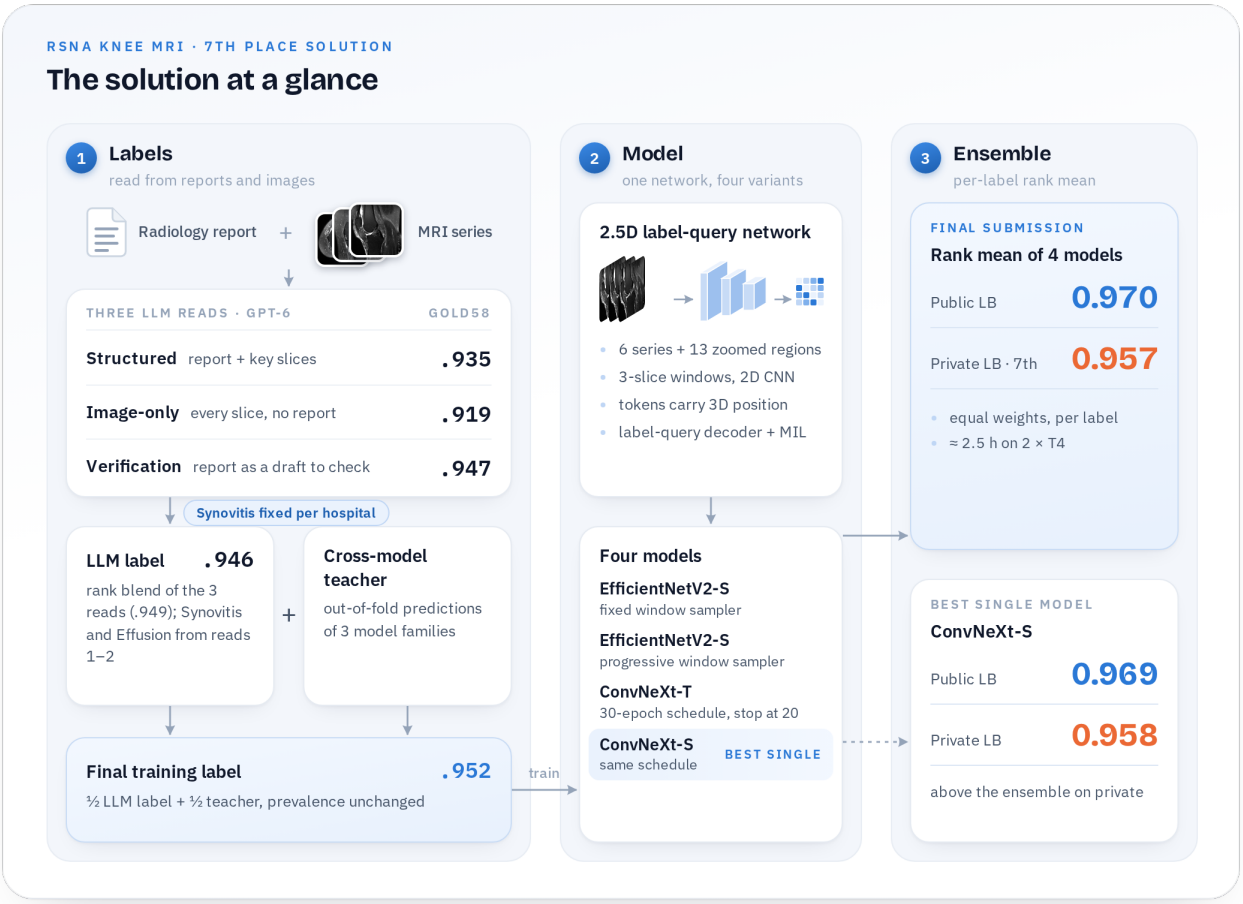
- Business context: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/overview
- Data context: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/data

Overview of the Approach

The test set has no reports, so a model can only be as good as the labels read out of the training reports. My solution has three parts:

1. **Labels.** GPT-6 astra reads each training report together with the MRI images, using three prompts. A per-hospital correction fixes Synovitis, and the result is blended with a cross-model teacher.
2. **Model.** One 2.5D label-query network, trained with four backbones/recipes.
3. **Ensemble.** A per-label rank mean of the four models.

Most of my gain came from the labels. The model itself is trained only on competition data: the competition images and labels extracted from the competition reports, starting from ImageNet-pretrained backbones.



Details of the Submission

1. Labels

The training set has 4,349 studies with free-text reports in 8+ languages and no image-level labels. Another 58 studies ("gold58") come with host labels; I used their host labels only to judge label quality.

Reports miss and hedge findings, and their wording depends on the hospital. Report-only LLM reads reached only .89–.91 macro AUC on gold58, so every read I use also looks at the images.

Three LLM reads. All three use GPT-6 astra with one shared per-finding protocol that defines what counts as positive for each of the 12 labels. Each read answers in JSON.

Read	Input	Prompt	gold58 AUC
Structured	report + key slices (9 sagittal, 9 coronal, 6 axial)	grade 24 attributes by the protocol, then give a probability per label	.935
Image-only	every slice as its own 384 px frame, no report	same protocol, from the images alone	.919
Verification	report + images	the report is a draft that may contain errors; check each finding against the images	.947

Blending the reads. The three reads are rank-blended per label, keeping each label's prevalence. The blend scores .949 on gold58, above every single read.

- For Synovitis and Effusion I keep the blend of reads 1–2 only. The verification read over-calls those two: it raises their prevalence from .24 to .31 and from .37 to .48.
- This lowers the gold58 score slightly, to .947. The result is the **LLM label**.
- The verification prompt was chosen on these 58 studies, so its .947 is optimistic.

Synovitis per-hospital correction. The LLM marks Synovitis almost exactly when the report uses the word (92% vs 6%). How often reports use the word depends on the hospital, not on the images. I use the scanner model as a stand-in for the hospital, at sites with ≥ 30 studies. Each site's Synovitis label gets one logit shift, solved so that the site's mean equals the rate expected from its image effusion level. The effusion level comes from an out-of-fold image model. The order of studies within a site does not change.

Cross-model teacher. The final label is ½ · LLM label + ½ · teacher.

- **What the teacher is.** The out-of-fold prediction of three 5-fold model families: two EfficientNetV2-S recipes and a ConvNeXt-T, each trained on a sibling LLM label. Each study is scored only by a model that never saw it.
- **How it is combined.** The teacher is rank-mapped onto the LLM label's values, so prevalence does not change.
- **Effect.** On gold58 it takes the label from .947 to .952. On the LB, the single-family teacher alone added +.003 public / +.002 private. Moving to the final recipe (three reads + three-family teacher) added another +.002 / +.004 for ConvNeXt-T (fig. 3).

Public labels. Every label version in this section is public in RSNA Knee LLM Labels, one file per step. Each file has one row per training study and the 12 labels as soft targets in [0, 1]; no report text is included.

File	What it is	gold58 AUC
`1_read_structured.csv`	Structured read: 12 probabilities + the 24 graded attributes	.935
`2_read_image_only.csv`	Image-only read	.919
`3_read_verification.csv`	Verification read (prompt chosen on gold58)	.947
`4_read_structured_synovitis_fixed.csv`	Structured read after the Synovitis correction, which is applied before blending	.937
`5_blend_reads_1_2.csv`	File 4 + image-only read, 60 : 40	.943
`6_blend_reads_1_2_3.csv`	File 5 + verification read, 1 : 1 rank mean	.949
`7_llm_label.csv`	**LLM label**: file 6 with Synovitis and Effusion from file 5	.947
`7a` / `7b` / `7c_llm_label_variant.csv`	Sibling LLM labels (other read weights), one per teacher family	.946 / .947 / .946
`8_final_label.csv`	**Final label**: ½ LLM label + ½ teacher; the four final models train on it	.952
`9_final_label_with_gold58_studies.csv`	File 8's recipe on the training set + the 58 gold studies, without host labels; the twins in final submission 2 train on it	–

2. Model



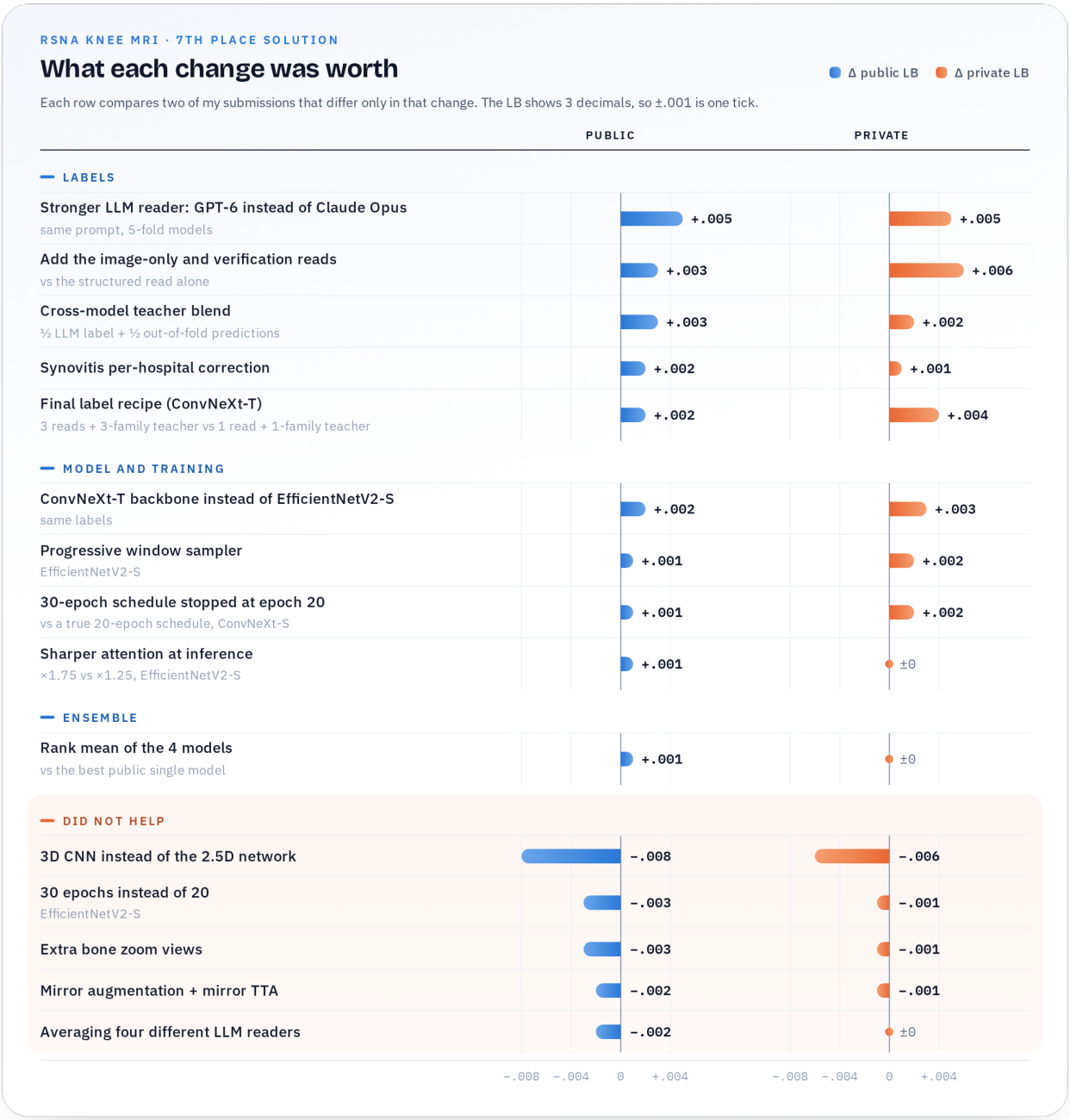
- **Input.**
  - Up to 6 series (sagittal, coronal and axial, each with and without fat suppression).
  - Each series is cropped to 140 mm at 384 px, right knees are mirrored, and up to 48 slices are kept per series.
  - A window is 3 adjacent slices.
  - 13 fixed zoomed regions (menisci, patellofemoral joint, compartments) are added as extra series.
- **Tokens.** A shared 2D CNN turns each window into 12×12 tokens. Each token carries its series and its 3D position in millimetres in a common knee frame, so all planes and crops line up.
- **Heads.** A label-query decoder, where 2 queries per label attend over every token of every window, and a gated attention-MIL head. The two heads are averaged.
- **Training.**
  - Soft-label BCE with positive weighting (PF OA ×2) and AdamW with a cosine schedule.
  - Weight averaging (SWA) of the last 5 epochs.
  - Affine and MRI-intensity augmentation, plus series dropout.
  - Every model trains on the full training set in about 6–10 h on a single GPU.
- **Inference.**
  - A window at every slice of every series and zoomed region.
  - Attention is sharpened for the much larger bag: ×1.75 for EfficientNetV2-S, ×1.25 for ConvNeXt.
  - fp16 on 2×T4, about 0.6 h per model.

3. Four models and the ensemble

Model	Recipe	Public	Private
EfficientNetV2-S	fixed number of windows per series, 20 epochs	.968	.954
EfficientNetV2-S	progressive window sampler	.969	.954
ConvNeXt-T	30-epoch schedule stopped at epoch 20	.969	.957
ConvNeXt-S	same as ConvNeXt-T	**.969**	**.958**
Rank mean of the 4	final submission 1	**.970**	**.957**
Rank mean of 8	the 4 + a twin of each; final submission 2	**.970**	**.957**

- **The ensemble.** An equal-weight, per-label percentile-rank mean of the four models. The whole kernel runs in about 2.5 h, well inside the 9 h limit.
- **Best single model.** ConvNeXt-S on its own already ranks first on the public LB (.969), and it scored .958 private, slightly above the ensemble.
- **Second pick.** My second selected submission added a twin of each model, trained with another seed on the training set plus the 58 gold studies (with my labels, not the host labels): 8 models in total. It scored the same as the four-model ensemble.

4. What worked and what didn't



Label changes gave the largest and most consistent gains, and they held up on the private LB.

Other things that did not help:

- multi-scale tokens;
- a slice-axis mixer;
- a larger decoder;
- multi-task loss balancing;
- heavier augmentation;
- larger crops.

Validation. Label quality was judged on the 58 gold studies. Every change in the chart above was checked on the leaderboard with a pair of submissions that differ only in that change.

Conclusion

In my experiments, changes to the labels moved the leaderboard more than changes to the model (fig. 3).

- **Image reads beat text reads.** Letting the LLM read the images next to the report, and asking it to verify the report rather than copy it, beat every text-only read.
- **Label noise had structure.** Some of it was systematic, such as the per-hospital Synovitis habit, and correcting that pattern helped more than the loss changes I tried.
- **The ensemble had little diversity.** The four models share the same labels, input pipeline and heads; only the backbone and training recipe differ. Their rank mean added +.001 public and nothing on private over the best single model, which most likely means they make similar mistakes. A more diverse ensemble (other architectures, inputs or label sources) is the clearest next step.

Acknowledgements

I am grateful to the Kaggle community for the open discussions and shared notebooks throughout the competition.

Sources

- Labels, every version above: https://www.kaggle.com/datasets/songqizhou/rsna-knee-llm-labels
- Inference notebook: https://www.kaggle.com/code/songqizhou/rsna-knee-7th-place-inference
- Training code: coming soon.
- Host ruling that LLM APIs may be used to extract labels from reports: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/discussion/733965
- Tan & Le, EfficientNetV2: Smaller Models and Faster Training, ICML 2021. https://arxiv.org/abs/2104.00298
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- Wightman, PyTorch Image Models (timm). https://github.com/huggingface/pytorch-image-models

Supplemental Q&A

Optimo · 2026-10-24T06:05:22.167000

Hey! Congratulations on an elegant solution! It does seem that distilling Astra was the key. Could you tell us a bit more about the 13 fixed region crops, how do you define them?

BarryZhou · 2026-10-24T14:32:34.743000

Thanks! Agreed, the labels did most of the work.

The crops are fixed boxes, the same in every study, with no localizer. Every slice is first centre-cropped to 140 mm (384 px) with a fixed orientation, and right knees are mirrored, so a box given as fractions of the slice lands on roughly the same anatomy in every study. The 13 boxes:

- **General (1.7× zoom):** the front of the axial FS slice (patellofemoral joint), the front of the sagittal FS slice (patella, suprapatellar recess, Hoffa's fat pad), and the medial and lateral halves of the coronal FS slice.
- **Menisci (2.2×):** the front and back halves of the tibiofemoral joint on sagittal FS, the back half on sagittal non-FS, and the medial and lateral joint lines on coronal FS.
- **Patellofemoral (~2×):** patella and trochlea on axial FS / non-FS and sagittal FS / non-FS.

Each crop is cut from its parent series' windows on the GPU and enlarged back to 384 px. It enters the same bag as extra windows, with its own view embedding and its true 3D position, so no separate head is needed. In training each box gets a random affine jitter, so it never has to sit exactly on the structure. Crops centred by a landmark localizer did not beat the fixed boxes.

Exact coordinates: `ROIS` in the inference notebook, https://www.kaggle.com/code/songqizhou/rsna-knee-7th-place-inference

Adel Zebiche · 2026-10-24T20:41:18.210000

Thanks for the very complete write-up! Labelling with MRI + report was clearly the key. Do you know roughly how much the (MRI + report) labelling pass cost in API fees (all ~4,400 exams at ~277 images each), and did you downsample the slices to keep it affordable?

BarryZhou · 2026-10-25T12:02:37.607000

Thanks! Here it is in tokens, so you can apply your own price. Per study (input + output):

- Structured read (report + 24 key slices): ~4.7k tokens
- Image-only read and verification read (every slice): ~20k tokens each

For all 4,349 studies, the three reads came to about 193 M input + 5 M output tokens, not counting prompt tests on the 58 gold studies.

On downsampling: only the structured read subsamples, using 9 sagittal / 9 coronal / 6 axial slices at 256 px, tiled into 3 images. The other two reads don't send all ~277 images either. They use one series per plane (fat-suppressed when available) and send every slice as its own 384 px frame of a 140 mm centre crop, about 92 frames per study. Series longer than 48 slices (4%) were evenly subsampled to 48. Each frame costs about 190 input tokens, so the images make up most of the bill.